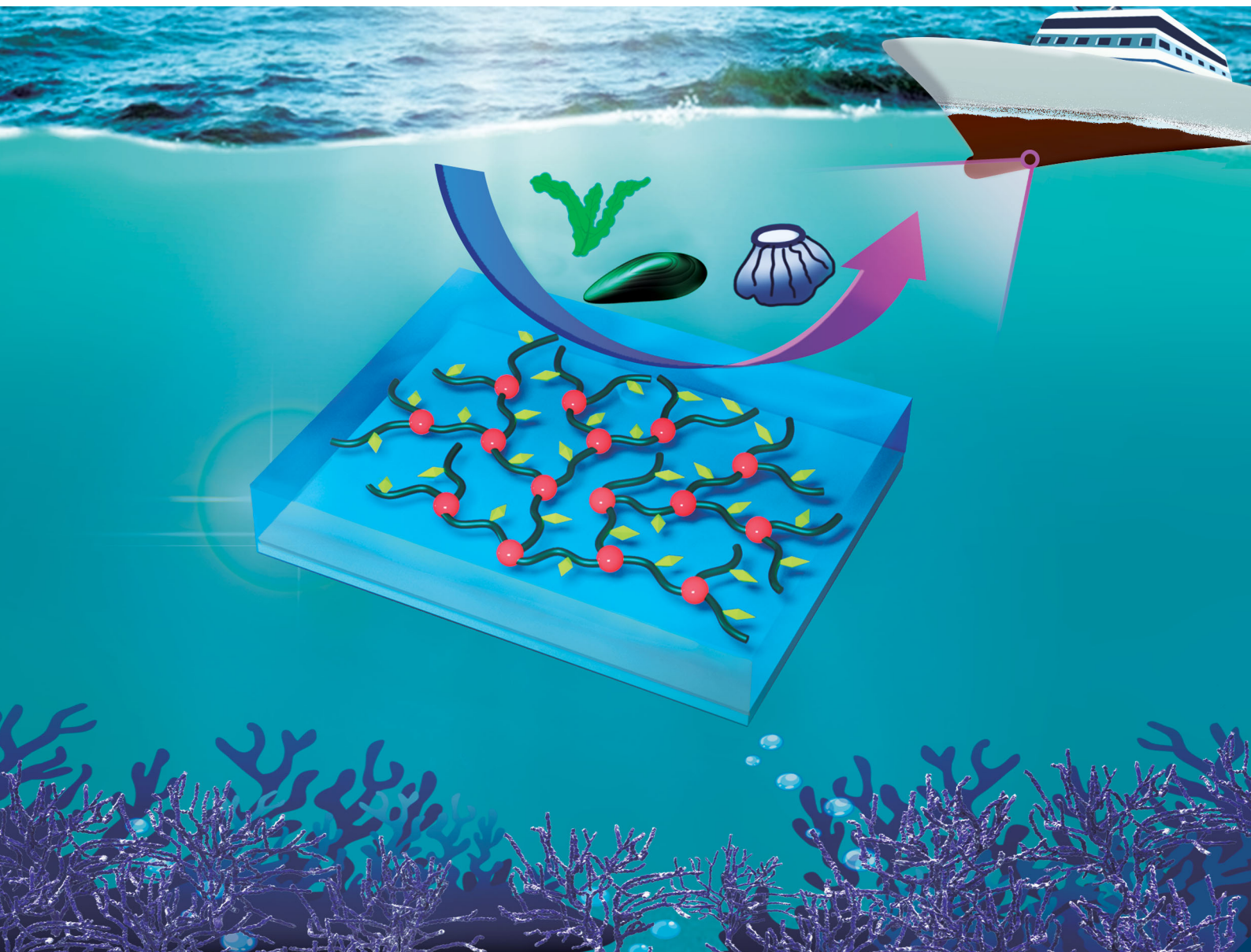


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PAPER

Chunfeng Ma *et al.*
Surface-fragmenting hyperbranched copolymers with
hydrolysis-generating zwitterions for antifouling coatings

PAPER

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8, 5434Surface-fragmenting hyperbranched copolymers
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antifouling coatings†Liqin Mei, Xiaoqing Ai, Chunfeng Ma * and Guangzhao Zhang 

Zwitterionic polymers have attracted increasing attention due to their excellent fouling resistance ability and eco-friendliness. Yet, their non-degradability and hydrophilic nature limit their applications. In this study, we have prepared a novel surface-fragmenting hyperbranched copolymer with tertiary carboxy-betaine ester (TCB) primary chains and poly(ϵ -caprolactone) (PCL) bridged chains, where the former and the latter can hydrolyze and degrade in marine environments, continuously generating zwitterions, so the polymer coating has a fouling resistant and renewable surface. Our study demonstrates that the degradation rate of the polymer is well controlled by the content of PCL bridges. Protein resistance and antibacterial assays show that the coating can inhibit the adhesion of protein and marine bacteria (*Pseudomonas* sp.). This new surface-fragmenting, self-regenerating hyperbranched zwitterionic copolymer has multiple applications in antifouling coatings.

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Introduction

Marine biofouling is a serious problem in marine industries and activities.^{1–3} The so-called self-polishing coatings (SPCs) or hydrolyzable acrylate polymers have been used in marine antibiofouling for decades, where the surface renewal due to the hydrolysis of polymers makes the release of the antifoulants possible.^{4–6} However, the surface renewal relies on a strong water flow, and the antifouling performance is poor under static conditions. Moreover, because the main chains of such polymers have a stable C–C structure, they cannot degrade in marine environments and this may cause environmental problems.^{7–9} Using biodegradable polymer based coatings can solve such problems since they can form a self-renewal surface by the cleavage of polymer main chains, which can occur even under static conditions. Besides, biodegradable polymers can finally degrade into small molecules and do not give rise to microplastic pollution. So, they have become the new-generation antifouling materials.^{10–13}

As a typical biodegradable polymer, poly(ϵ -caprolactone) (PCL) is particularly attractive because of its excellent biodegradability and biocompatibility.^{14–16} However, it cannot be used directly in marine antibiofouling because of its slow degradation and poor adhesion strength. Generally, it needs

chemical or physical modification.^{17,18} Recently, biodegradable polyacrylates with the advantages of both polyacrylates and biodegradable polymers have been developed for marine antibiofouling. They not only possess controlled degradation but also display remarkable film-formation performance as well as adhesion to various substrates.^{19–21} Yet, the coating with the linear polymer generally has a high viscosity, limiting applications. Hyperbranched polymers (HBPs) are well known to have lower chain entanglement, lower crystallinity, higher solubility, and greater end-up functionalities as compared with their linear analogues.^{22–24} Particularly, hyperbranched biodegradable polyacrylates can be readily synthesized by one-step reversible addition-fragmentation chain transfer (RAFT) polymerization and self-condensing vinyl polymerization.^{25–28}

Note that the antifouling ability of biodegradable polymers increases with the degradation rate. However, a high degradation rate leads inevitably to a short service life.^{29,30} To solve the contradiction, antifoulants are generally added to biodegradable polymers, which may cause ecological problems. Therefore, it is necessary to incorporate fouling resistant groups into the degradable polymers to improve their antifouling duration. Zwitterions can effectively inhibit protein adsorption and microbial adhesion, and thus they are likely good candidates.^{31,32} Unfortunately, zwitterions have a hydrophilic nature and their presence usually causes the coating to swell with poor mechanical properties.³³ Another problem is that zwitterionic polymers cannot resist inorganics or dead microorganisms in complex marine environments, which could cover the surfaces, leading to a failure of antifouling function. Moreover, it is challenging to incorporate

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polar zwitterions into a hydrophobic biodegradable polymer because they seldom have a co-solvent.³⁴

In this study, we have synthesized a hyperbranched copolymer *via* RAFT polymerization of divinyl-functionalized poly(ϵ -caprolactone) (PCL- V_2), tertiary carboxybetaine ester (TCB), and methyl methacrylate (MMA). The hydrolysis of TCB segments can generate zwitterions, which make the surface resistant to fouling. The introduction of branching can reduce the crystallinity and improve the degradation of PCL bridges, which makes the surface renewable. Thus, this would create a fragmenting surface that enables the coating to continuously self-regenerate zwitterions. The novel combination of self-renewable polymers and zwitterionic polymers would show a synergistic effect and solve some challenging problems such as the contradiction between antifouling ability and service life of marine antifouling coatings or the non-persistent antifouling ability of traditional zwitterionic polymers. We have investigated the degradation, protein resistance, and antibacterial properties of the polymers. Our aim is to develop an eco-friendly and highly effective antifouling system.

Experimental

Materials

Methyl methacrylate (MMA, 99%, Aladdin) was purified by distillation before use. 2,2-Azobisisobutyronitrile (AIBN, 98%, Aladdin) was recrystallized twice from methanol. Tetrahydrofuran (THF, Titan, AR) and toluene (Guangzhou Chemical Reagent, AR) were refluxed using CaH_2 and distilled before use. Hydroxyl end-functionalized poly(ϵ -caprolactone) diol (PCL, $M_w \approx 2000 \text{ g mol}^{-1}$) was purchased from Perstorp, and 2-cyano-2-propyl dodecyl trithiocarbonate (CPDT, 95%) was purchased from Zhengzhou JACS Chem Co., Ltd. Ethyl acrylate (99%), 2-methylaminoethanol (99%), acryloyl chloride (96%), triethylamine (TEA, 99%), and 2-isocyanatoethyl methacrylate (IMA, 98%) were purchased from Aladdin Chemistry Co., Ltd. Anhydrous dimethylformamide (DMF, 99.9%) was purchased from Energy Chemical Co., Ltd. Tertiary carboxybetaine ester (TCB) was synthesized according to a previously reported procedure (Fig. S1, ESI†).³⁵ Artificial seawater (ASW) was prepared following the ASTM D1141-98 (2013) standard. Other reagents were used as received.

Synthesis of divinyl-functionalized poly(ϵ -caprolactone) (PCL- V_2)

PCL- V_2 was synthesized according to a procedure described elsewhere (Scheme 1a).^{15,36} Typically, PCL diol (20.00 g, 0.01 mol), IMA (3.10 g, 0.02 mol), and toluene (20 mL) were added into a two-neck round-bottom flask with a magnetic stirring bar. After the solution became homogeneous, three drops of TEA were added as a catalyst. The reaction was performed in a nitrogen atmosphere at 80 °C for 24 h. Then, the toluene was completely removed under vacuum. ^1H NMR (600 MHz, CDCl_3 , ppm, Fig. S2, ESI†): 6.10, 5.59 ($\text{CH}_2\text{C}(\text{CH}_3)$), 4.21 ($\text{COOCH}_2\text{CH}_2\text{NHCO}$), 4.04 ($\text{OCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CO}$),

3.86 ($\text{OCH}_2\text{C}(\text{CH}_3)_2\text{CH}_2\text{O}$), 3.45 ($\text{COOCH}_2\text{CH}_2\text{NHCO}$), 2.28 ($\text{OCH}_2\text{-CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CO}$), 1.92 ($\text{CH}_2\text{C}(\text{CH}_3)$), 1.68–1.58 ($\text{OCH}_2\text{CH}_2\text{CH}_2\text{-CH}_2\text{CH}_2\text{CO}$), 1.36 ($\text{OCH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CO}$), 0.95 ($\text{OCH}_2\text{-C}(\text{CH}_3)_2\text{CH}_2\text{O}$).

Synthesis of the hyperbranched copolymer

The hyperbranched copolymer was prepared by RAFT polymerization of MMA, TCB, and PCL- V_2 (Scheme 1b). Typically, a mixture of MMA (0.46 g, 4.60 mmol), TCB (1.16 g, 5.06 mmol), PCL- V_2 (0.69 g, 0.30 mmol), CPDT (86.4 mg, 0.25 mmol), and AIBN (8.2 mg, 0.05 mmol) was dissolved in anhydrous DMF (6.00 g) and added to a 25 mL Schlenk flask with a magnetic stirring bar. Then, the mixture was degassed by using three freeze–evacuate–thaw cycles and was backfilled with nitrogen. The reaction flask was placed in an oil bath at 70 °C for 24 h. The polymerization was quenched by rapid cooling in liquid nitrogen and exposure to air. The polymer was precipitated in ether and dried overnight under vacuum at 60 °C. For all polymerizations, the weight ratio of TCB was invariable, and the molar ratio of [CDPT]:[AIBN] was maintained at 1:0.2, while the feed ratio of [MMA + TCB]:[PCL- V_2] was varied from 40:0.3 to 40:4.0. The characterization data of the hyperbranched copolymers are shown in Table 1. For convenience, the hyperbranched copolymer containing 22 wt% MMA, 46 wt% TCB, and 32 wt% PCL- V_2 was named *h*-PCL32. Similarly, the linear copolymer containing MMA and TCB was named LMT.

Characterization

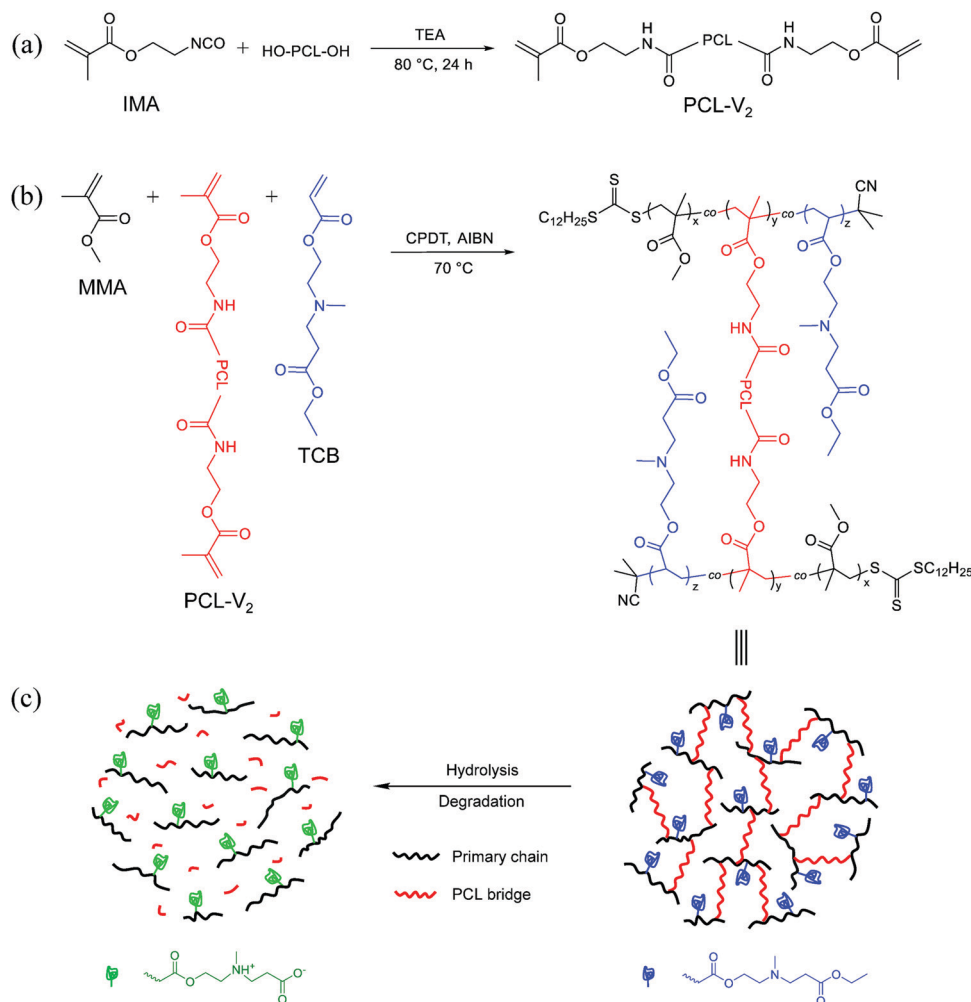
The structures of the hyperbranched copolymers were analyzed using ^1H and ^{13}C nuclear magnetic resonance (NMR) spectroscopy (Bruker, 600 MHz) and Fourier-transform infrared (FTIR) spectroscopy (Bruker VERTEX 70). The number-average molar mass (M_n) and polydispersity (D_M) were determined at 35 °C by gel permeation chromatography (GPC) on an instrument equipped with refractive index and ultraviolet detectors. THF was used as the eluent at the flow rate of 1.0 mL min^{-1} . A series of narrowly dispersed polystyrene (1.30×10^3 to $2.21 \times 10^6 \text{ g mol}^{-1}$) standards was used for calibration.

Differential scanning calorimetry (DSC)

The crystallization temperature (T_c) was determined by a DSC test (NETZSCH DSC 204F1) under a nitrogen flow of 50 mL min^{-1} . Subsequently, a sample was quickly heated to 120 °C and balanced for 5 min to remove the thermal history. Then, it was cooled to -50 °C at a rate of 10 °C min^{-1} . Subsequently, it was heated to 120 °C at a rate of 10 °C min^{-1} . T_c was obtained from the endothermic shift in the second heating scan.

Quartz crystal microbalance with dissipation (QCM-D)

The enzymatic degradation and protein adsorption were measured by QCM-D using an instrument from Q-sense AB (Sweden). The films were prepared by spin-casting on an AT-cut quartz crystal. After immersion in 0.1 M NaOH solution for 3 h at 25 °C, the sample was hydrolyzed and taken out; the sample was rinsed with deionized water and dried under nitrogen flow.



Scheme 1 (a) Synthesis of PCL-V₂; (b) synthesis of hyperbranched copolymers by RAFT polymerization; and (c) hydrolysis and degradation of hyperbranched copolymers.

Table 1 Characterization data of hyperbranched copolymers

Sample	Feed molar ratio (MMA + TCB): PCL-V ₂ :CPDT:AIBN	MMA/TCB ^a (wt%)	PCL ^a (wt%)	M_n^b (kg mol ⁻¹)	D_M^b	DB ^c (mol%)
LMT	40/0/1/0.2	54/46	0	10.0	1.89	—
<i>h</i> -PCL11	40/0.3/1/0.2	45/44	11	15.5	1.95	0.7
<i>h</i> -PCL32	40/1.2/1/0.2	22/46	32	18.7	1.96	2.7
<i>h</i> -PCL53	40/4.0/1/0.2	0/47	53	17.4	3.18	4.6

^a Weight ratio was determined by ¹H NMR. ^b Determined by GPC. ^c Degree of branching (DB) was calculated according to the equation: DB (mol%) = $(A_b / (A_b + A_{I-MMA} + A_{I-TCB})) \times 100$, where A_b represents the integral area of branched PCL units, A_{I-MMA} represents the integral area of linear MMA units, and A_{I-TCB} represents the integral area of linear TCB units in hyperbranched copolymers determined by ¹H NMR.

Subsequently, the quartz crystal was mounted in a fluid cell with the sample side exposed to the solution. Fibrinogen solution (1.0 mg mL⁻¹) or lipase PS solution (1.0 mg mL⁻¹) was added to the surface at a flow rate of 150 $\mu\text{L min}^{-1}$ using ASW as the reference. The details can be found elsewhere.³⁷

The change in the frequency shift (Δf) of the crystal is related to the mass change of the test layer on the quartz crystal, whereas the change in the dissipation shift (ΔD) is related to the viscoelastic properties of the additional layer. Therefore, the changes in Δf and ΔD reflect the protein adsorption and the

structural change of the films, respectively. All experiments were carried out at 25 $^{\circ}\text{C}$, and the data were obtained from the third overtone ($n = 3$).

Attenuated total reflection infrared (ATR-IR) spectroscopy

The non-hydrolyzed films were recorded by using a Bruker VERTEX 70 infrared spectrometer. The resolution was 4 cm⁻¹, and the accumulation was 32 scans. After immersion in 0.1 M NaOH solution for 3 h at 25 $^{\circ}\text{C}$, the ATR-IR spectrum of the hydrolyzed films was measured using the same method.

Water contact angle (WCA) measurements

The WCA measurements were performed using a Theta Auto 113 (KSV NIMA, Biolin) at 25 °C by dropping 3 μL of deionized water droplets on the sample surface using the sessile method. After immersion in 0.1 M NaOH solution for 3 h, the WCA of the hydrolyzed sample was measured using the same method. Five regions were tested for each sample for obtaining the average value.

Adhesion test

The adhesion strength of the non-hydrolyzed coatings to the epoxy panel substrate was measured by using an adhesion tester (PosiTest AT-A Automatic) according to the ASTM D4541-09. The data were obtained by detaching an aluminum dolly (20 mm in diameter) at a speed of 0.2 MPa s^{-1} . After immersion in 0.1 M NaOH solution for 3 h at 25 °C, the adhesion strength of the hydrolyzed coatings was measured using the same method. Five regions were tested for each sample to obtain the average value.

Antibacterial assays

The marine bacterial species *Pseudomonas* sp. NOV776 was used to evaluate the antibacterial ability following our previous procedure.¹⁹ The film on a silicon slide (10 $\times$ 10 mm²) was prepared by spin-casting and was hydrolyzed in 0.1 M NaOH solution for 3 h before use. The bacterial concentration was adjusted to 10⁸ cells per mL using ASW. Each sample was immersed in a bacterial suspension (1 mL) for 4 h. Subsequently, it was washed gently with ASW and stained by using the LIVE/DEAD BacLight Bacterial Viability Kit. A fluorescence microscope (Scope A1, Zeiss) was used to observe the adhered bacterial cells. The relative bacterial adsorption (RBA) was calculated by using the ImageJ software, in which the control was the average number of bacteria on the silicon wafer.

Results and discussion

To ensure that the hyperbranched copolymers have surface-fragmenting and fouling resistant abilities, the primary chain should not be too long, and the hydrolyzed products should have antifouling functions. Otherwise, after the PCL bridges were completely degraded, the primary chains would not be polished from the surface, which would result in poor anti-fouling effects. We used RAFT polymerization because of its wide range of monomers, adjustable polymer structure, and facile polymerization procedure.²⁵

In addition, to reduce the crystallinity and improve the degradability, the molecular weight of the PCL should not be too high. We selected PCL oligomers as the branched chains. Fig. 1a shows the ¹H NMR spectra of LMT and *h*-PCL32. The peak at 3.70–3.50 ppm for LMT is attributed to the protons of the CH₃ group of MMA. The peaks at 4.30–4.10, 2.85–2.20, and 1.27–1.19 ppm can be assigned to TCB. For *h*-PCL32, all the peaks of MMA and TCB can be clearly observed. Moreover, the new peaks at 4.04, 3.86, 3.42, 1.68–1.58, 1.36, and 0.95 ppm

were attributed to the PCL branched chains. Furthermore, as shown in the ¹³C NMR spectra (Fig. S3, ESI[†]), we can observe that the new peaks from the PCL branched chains appear for *h*-PCL32 besides the peaks from MMA and TCB. These results show that the successfully synthesized hyperbranched copolymer consisted of TCB primary polymer chains and PCL bridged chains. The degree of branching (DB) can be calculated from the ¹H NMR spectra according to a procedure described elsewhere.³⁸ Table 1 shows that all the samples have similar TCB contents but different PCL contents, and when [PCL-V₂] was increased from 0.3 to 4.0, the DB of *h*-PCL_x gradually increased from 0.7 to 4.6, which indicated the presence of branched structures in the copolymers. Fig. 1b shows the FTIR spectra of LMT and *h*-PCL32. Compared with LMT, a new peak at 1525 cm⁻¹ appears for *h*-PCL32, which is attributed to the C–N stretching and N–H bending of the urethane group from PCL-V₂. The result further indicates that the copolymers contain PCL fragments.

As we know, crystallinity profoundly affects the degradation. We investigated the effects of PCL bridges on the crystallinity of the copolymers. Fig. 2 shows the DSC curves for PCL-V₂ and *h*-PCL_x. The crystallization temperature (*T*_c) of PCL-V₂ and *h*-PCL53 was clearly observed to be 45.0 °C and 42.1 °C, respectively. *T*_c of *h*-PCL53 is high because it has a high content of PCL fragments. Interestingly, for *h*-PCL11 and *h*-PCL32, we did not observe the melt peak in the range from –50 °C to 120 °C. This fact indicates that the copolymerization reduces the PCL crystallinity, which would make the copolymer more degradable.

Fig. 3 shows the time dependence of the frequency shift (Δf) and the dissipation shift (ΔD) for the enzymatic degradation of the polymers investigated by using QCM-D. It is known that an increase in the mass of the test layer causes Δf to decrease.²⁹ For LMT, we observed an obvious decrease in Δf after the introduction of lipase PS, indicating that lipase PS had been adsorbed on the LMT film. This is understandable because the copolymer did not have any degradable PCL fragments. For *h*-PCL11, we can observe a decrease in Δf relative to the baseline. After rinsing with ASW, *h*-PCL11 increased in Δf relative to the baseline, indicating that the contribution to the mass by the adsorption of lipase PS is less than that due to degradation. For *h*-PCL32 and *h*-PCL53 with high PCL contents, Δf shows a remarkable increase relative to the baseline, indicating the mass loss of the films. The degradation of PCL bridges generates the copolymer fragments, which subsequently disperse into the solution. Clearly, the degradation rate increases with the content of PCL bridges.

On the other hand, ΔD is related to the viscoelastic properties of the additional layer and reflects the structural change of the films.²⁹ The increase in ΔD relative to the baseline for LMT further indicates the formation of a viscoelastic layer of lipase PS. For *h*-PCL11, the increase in ΔD is less than that in LMT because the degradation of the PCL bridges creates a thin layer. Similarly, *h*-PCL32 has a relatively smaller increase in ΔD because it forms a thinner layer with quicker degradation. Note that the increase in ΔD for *h*-PCL53 is larger than those for

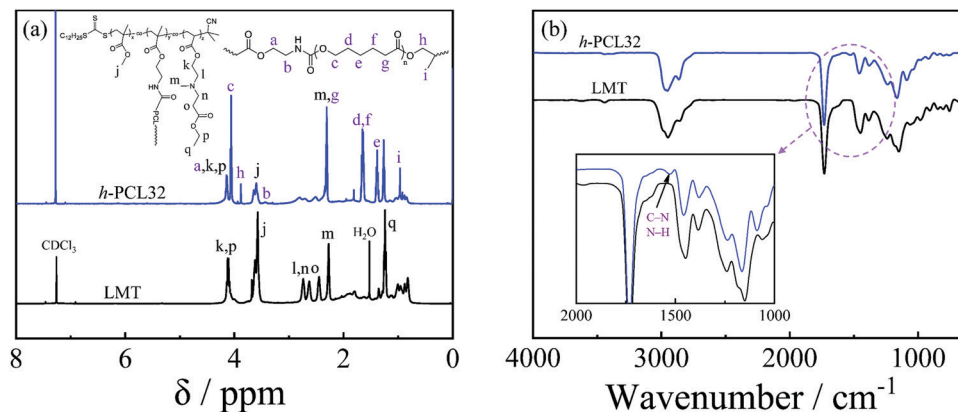


Fig. 1 (a) ^1H NMR and (b) FTIR spectra of copolymers. The inset shows the enlargement of the area between 1000 and 2000 cm^{-1} from the FTIR spectra.

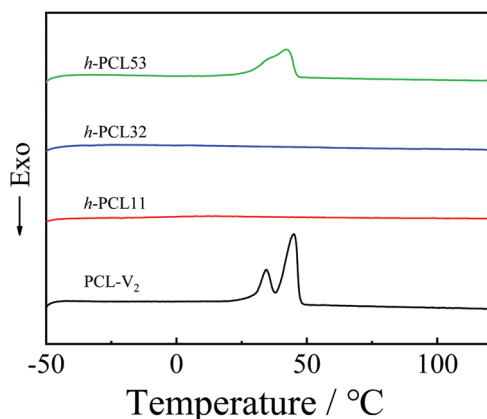


Fig. 2 DSC curves of the copolymers.

h-PCL11 and *h*-PCL32 even though it has the largest degradation rate among these copolymers. As we know, the degradation of the polymer is affected by its crystallinity and the content of cleavable linkage.¹⁸ As shown in Fig. 2, the DSC curve clearly indicates that *h*-PCL53 is a crystalline polymer. The degradation occurs preferentially in the amorphous regions and then in the crystalline areas. In other words, the degradation is heterogeneous, which leads to an uneven surface with a large ΔD .³⁹ However, the QCM-D measurements clearly show that the degradation rate increases with the content of PCL bridges.

We examined the hydrolysis-generating zwitterions and their effect on the wettability of the coating using *h*-PCL32 as a representative since all the samples have similar TCB contents (Table 1). Fig. 4a shows the ATR-IR spectra of *h*-PCL32 before and after hydrolysis. After hydrolysis, new bands can be observed at 1560 cm^{-1} , indicating that TCB is hydrolyzed into carboxybetaine groups. Furthermore, the band at 1645 cm^{-1} and an additional broad shoulder at 3420 cm^{-1} are attributed to OH from the hydration of the zwitterions. Fig. 4b shows the water contact angle (WCA) values of *h*-PCL32 before and after hydrolysis. The initial WCA of *h*-PCL32 is approximately 96° , indicating that the surface is hydrophobic. After hydrolysis, the WCA decreases to approximately 86° , indicating that the

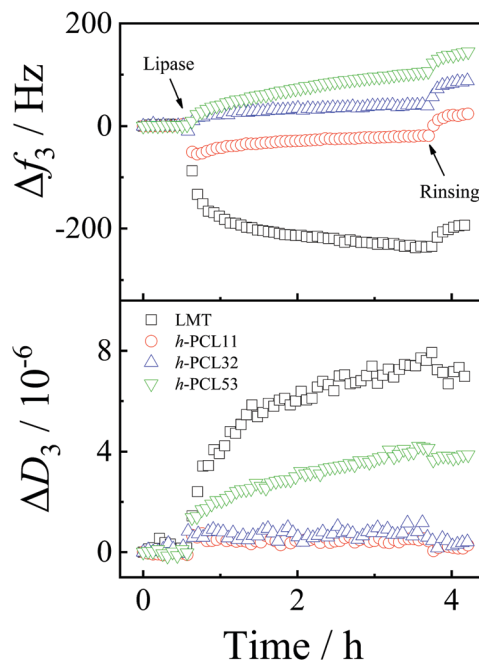


Fig. 3 Time dependence of frequency shift (Δf) and dissipation shift (ΔD) for the enzymatic degradation of copolymers in ASW at 25°C .

surface becomes hydrophilic because the hydrolysis of the TCB groups can generate zwitterions on the surface.

We also examined the effect of hydrolysis on the adhesion strength since it is essential for antifouling coatings (Fig. 5). Before and after hydrolysis, the adhesion strength of *h*-PCL32 was slightly changed (by approximately 2.0 MPa), indicating that the coatings have good adhesion to substrates and can be used in harsh marine environments. This is because TCB hydrolysis occurred only on the surface, and the matrix was not affected. Moreover, the degradable PCL bridges facilitate the removal of the hydrophilic surface.

We investigated the antifouling performance of the copolymer by using protein and marine bacteria as the models. Fig. 6a shows the time dependence of the frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of fibrinogen on

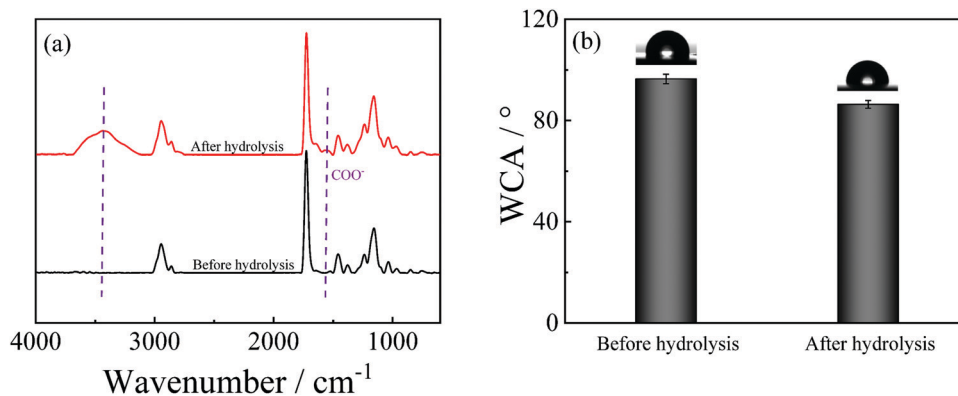


Fig. 4 (a) ATR-IR spectra and (b) water contact angles (WCAs) of *h*-PCL32 before and after hydrolysis.

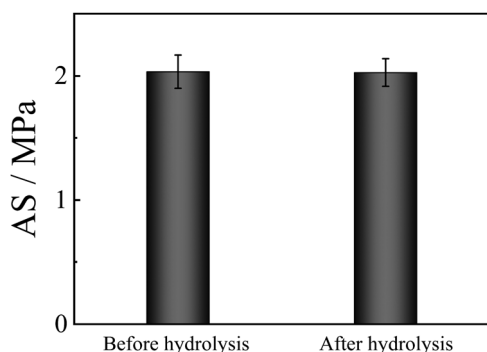


Fig. 5 Adhesion strength of *h*-PCL32 before and after hydrolysis.

h-PCL32 measured by QCM-D. Fibrinogen is a hydrophobic protein with a high molecular weight and is easily adsorbed onto hydrophobic surfaces.⁴⁰ For *h*-PCL32, before hydrolysis, Δf decreases and ΔD increases sharply once fibrinogen is

introduced, and then the values gradually level off. After rinsing with ASW, Δf and ΔD exhibit large changes relative to their baseline values. As we know, Δf decreases as the layer mass increases, whereas ΔD increases with the thickness and looseness of the test layer.^{29,39} Clearly, the fibrinogen adsorbs on *h*-PCL32 and forms a viscoelastic layer. For *h*-PCL32 after hydrolysis, Δf and ΔD after rinsing were close to the baseline, indicating almost no fibrinogen adsorption. As discussed above, the hydrolysis of TCB generates zwitterions forming a hydrophilic surface with protein resistance. Actually, we also examined the protein resistance of the LMT, *h*-PCL11 and *h*-PCL53 coatings after hydrolysis, which exhibited competitive antifouling performance in comparison with the *h*-PCL32 coating due to the similar contents of TCB. Fig. 6b shows the fluorescence images of *Pseudomonas* sp. and the relative bacterial adhesion (RBA) on different surfaces. A large number of bacteria can be observed on the silicon wafer (RBA ~ 100%), indicating that the fouling pressure was high. For *h*-PCL32,

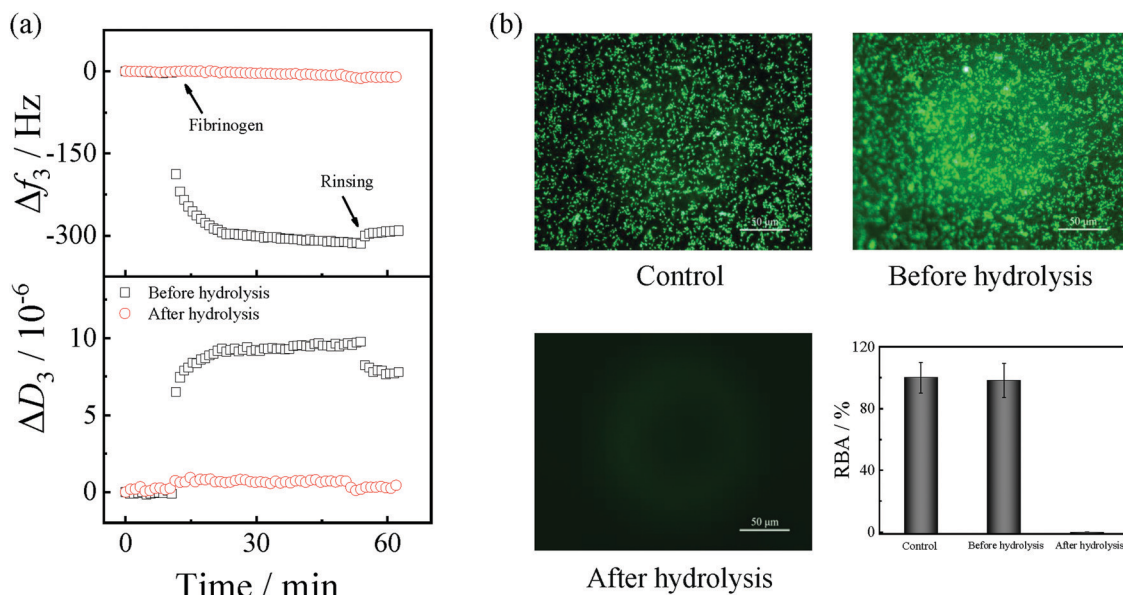


Fig. 6 (a) Time dependence of frequency shift (Δf) and dissipation shift (ΔD) for the adsorption of fibrinogen on *h*-PCL32. (b) Fluorescence images of *Pseudomonas* sp. adhered on different surfaces and the relative bacterial adhesion (RBA).

before hydrolysis, a large number of bacteria (RBA $\sim$ 98%) were observed because of the lack of antifouling moieties on the surface. After hydrolysis, almost no bacteria were observed. This fact further indicates that the hydrolysis of TCB effectively resists the adhesion of bacteria.

Conclusions

We prepared a novel surface-fragmenting hyperbranched copolymer with hydrolysis-generating zwitterions *via* RAFT polymerization, in which TCB primary chains were bridged with PCL segments. The degradation of PCL bridges and the hydrolysis of TCB created a dynamic surface, making the coatings fouling resistant and self-renewable. The degradation rate of the polymer can be well controlled by the content of the PCL bridges. Moreover, the polymer exhibited excellent protein resistance and antibacterial performance because of the zwitterions generated from the hydrolysis of the side groups. Such a hyperbranched copolymer can find applications in antifouling coatings.

Conflicts of interest

There are no conflicts to declare.

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